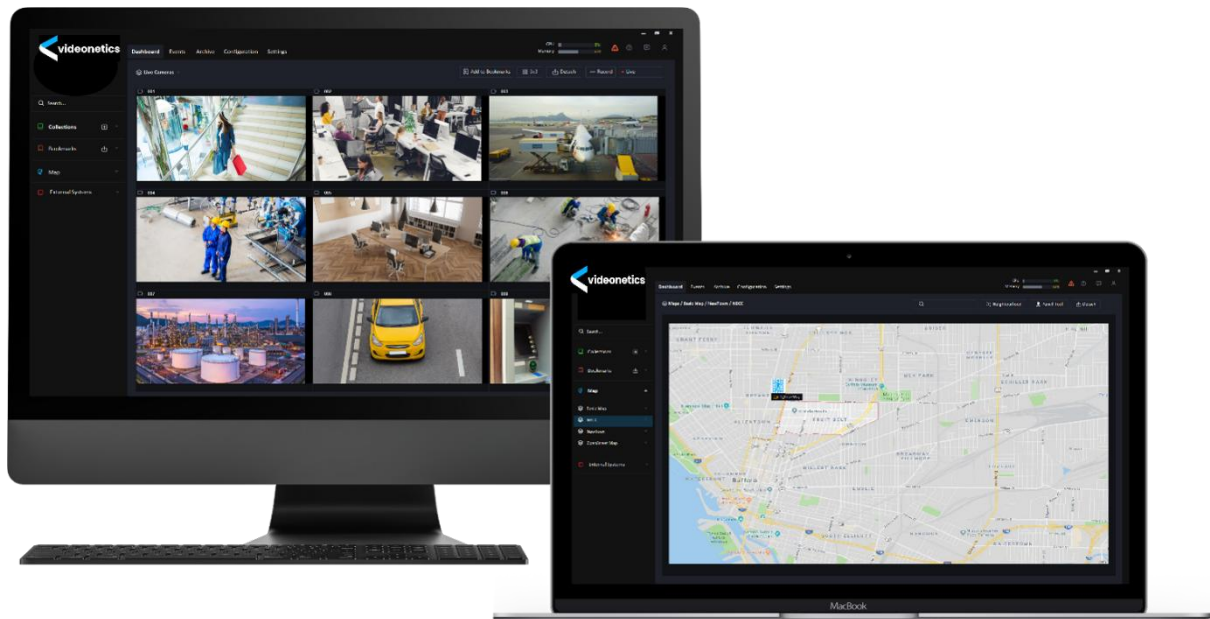


Videonetics

Job Description – Junior Software Engineer



Job Description: Junior Software Engineer

Location: Kolkata

Department: Technology

Mode of Employment – Full-time

About Videonetics:

Videonetics is a leading innovator in AI-powered video computing solutions, offering intelligent video management, analytics and security solutions across industries. Our mission is to make the world smarter and safer through cutting-edge technology.

To know more about us – <https://www.videonetics.com>, [videonetics: Overview | LinkedIn](#)

Who You'll Work With

As part of our Technology team, you will collaborate with a diverse and cross-functional group of professionals including machine learning engineers, computer vision researchers, embedded software engineers, systems architects and data engineers. You will work closely with domain experts and integration teams to translate real-world challenges into scalable AI-driven solutions. Your role will contribute to both research and production environments, supporting the development, optimization and deployment of AI models within real-time video pipelines and edge-based systems.

What you'll do

- Design and develop high-performance software solutions for computer vision applications using C++ and Python.
- Work with GStreamer or similar video frameworks to embed AI inference in real-time video pipelines.
- Work on RTP/RTSP-based streaming, addressing synchronization challenges (PTS/DTS), jitter management and latency control.
- Port and optimize AI models for computer vision tasks to run efficiently on edge and embedded platforms using frameworks like TensorRT, ONNX Runtime, or OpenVINO.
- Collaborate with video pipeline teams to ensure models are compatible with real-time inference requirements (e.g., for NVIDIA DeepStream or TensorRT deployment).
- Build and extend real-time video streaming workflows using GStreamer, ensuring optimal GPU utilization and low-latency performance.
- Perform data analysis, visualization and curation to support training on high-quality and diverse datasets.
- Contribute to both academic publication efforts and applied R&D initiatives.
- Collaborate with DevOps to containerize video applications using Docker and deploy in Linux-based environments.

Key Responsibilities:

- Design, develop and optimize software solutions in C++ and Python for high-performance Computer Vision Products.
- Implement and optimize data structures and algorithms to ensure efficiency and scalability.
- Manage memory, concurrency and resource usage across heterogeneous compute environments (CPU/GPU).
- Develop and maintain custom GStreamer components for real-time video acquisition, processing and output.
- Implement and debug streaming logic using RTP, managing PTS/DTS synchronization effectively.
- Design and implement scalable training and validation workflows for large-scale datasets.
- Ensure stream quality, low latency and resilience under different network conditions.
- Tune model architectures and hyperparameters for optimal accuracy and inference speed and evaluate model performance
- Collaborate with engineering teams to optimize models for deployment in edge environments or real-time systems.
- Maintain detailed documentation of experiments, training results and model behavior.
- Contribute to internal tools, utilities and reusable components for efficient experimentation and deployment.
- Support knowledge sharing through papers, or patents.

What we are looking for:

- B.E./B.Tech in Computer Science and Engineering.
- M.E./M.Tech in Computer Science and Engineering with B.E./B.Tech in Computer Science and Engineering.
- Strong foundation in algorithms, data structures and computer networks.
- Excellent programming skills in C/C++ and Python, with the ability to write clean, efficient and well-structured code.
- Good understanding of operating systems and experience in Linux system programming.
- Good understanding of mathematical fundamentals, including linear algebra, calculus, probability and statistics.
- Strong analytical and problem-solving skills, curiosity, and a willingness to learn new technologies and research methods.
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Why Join Us?

- Be part of an innovative company at the forefront of AI-driven video computing.
- Opportunity to work with top-tier partners and industry leaders.
- Competitive salary, performance incentives and professional growth opportunities.
- Embark on an impactful journey with Videonetics, where you'll work on innovative products that enhance safety, efficiency and sustainability, ultimately making a positive difference in society
- Global presence: Collaborate with international clients and teams, expanding your professional horizons

If you are passionate about building strong Product and driving strategic decisions, we invite you to be part of our growth journey at Videonetics!



We look deeper to simplify the world.

Making it

Safer. Smarter. Happier.

India | Singapore | Dubai

Corporate Office

Plot No AI/154/1, Action Area – 1A,
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Kolkata – 700156, India.

For More Information

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Visit us: www.videonetics.com

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